# NETWORK SUPERVISION WITH NAGIOS CORE

## RUNNING UBUNTU SERVER


### Summary

Mise en place d'une solution de monitoring centralisée sur Ubuntu Server pour surveiller en temps réel la disponibilité des infrastructures et des services réseau. Ce projet inclut la configuration d'alertes automatiques et l'utilisation de l'agent NRPE pour assurer la continuité de service des machines distantes.



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# Introduction :

## 1. Contexte et justification du sujet

With the rapid advancement of technology in an era where digital transformation is moving at a giant pace, computer networks have become essential pillars for ensuring service availability, security, and performance. Their proper functioning directly impacts productivity and business continuity. Network monitoring, a technical discipline that involves real-time tracking of the status and performance of IT equipment, has therefore become a necessity to guarantee the quality of service expected by end users.

## 2. Problématique

Given the major challenge of orchestrating data flow across the network, a central problem arises: how to monitor network status in real time, anticipate failures, and guarantee high service availability, all while transforming this monitoring into added value for the organization.

## 3. Hypothèse

If an open-source network monitoring solution is correctly deployed and configured, it will enable effective, real-time surveillance of the IT infrastructure, thereby reducing the risk of downtime and improving responsiveness to anomalies without significantly impacting the organization's revenue or operations.

## 4. Objectif général

To implement a reliable and scalable network monitoring solution based on Linux to ensure the continuous surveillance of critical IT network equipment and services.

# Chapter 2: Presentation, interpretation and validation of results

## Section 1: Solution Presentation and Environment Preparation

### I. VMware Workstation 17 pro Overview

**What is VMware Workstation?**

VMware Workstation is a Type 2 hypervisor, meaning it is a software application that allows you to create and manage one or more virtual machines (VMs) on a single physical computer (the "host"), whether it is running Windows or Linux.

It is a widely used tool in the fields of software development, testing, training, and the virtualization of complex IT environments.

### II. Presentation of the operating systems (Windows server 2022 and Ubuntu server), Nagios core and Pfsense.

#### 1. Presentation of the operating systems:

First of all, an **operating system** is the core software that controls all of the computer's **hardware** and manages the execution of other software (**applications**).

It acts as an intermediary between:

- **The hardware** (processor, memory, hard drive, network cards, peripherals, etc.)
- **The users** (you)
- **The software** (web browser, word processor, web server, etc.)

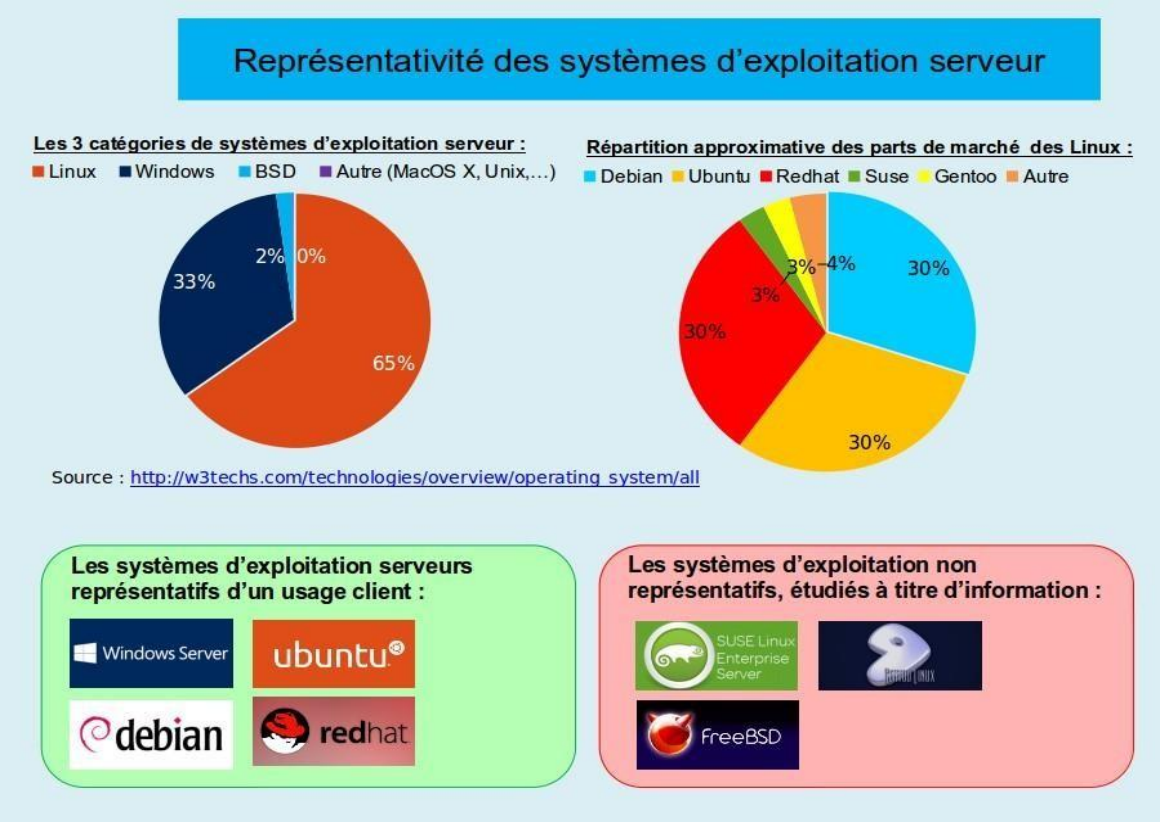


*Figure 3: Representativeness of operating systems*

- **Windows server :**

**Windows Server 2022** is Microsoft's server edition released on August 18, 2021, succeeding Windows Server 2019 and preceding Windows Server 2025. It offers enhanced security, high performance, deep cloud integration, and modern management tools.

It is designed for demanding and hybrid infrastructures, including data centers, file/DHCP/AD servers, and virtualized or containerized environments.

- **Ubuntu server :**

**Ubuntu Server** is a free and open-source Linux distribution renowned for its stability and ease of use. Based on Debian, it provides a modern server platform with a regular release cycle (every 6 months) and **Long Term Support (LTS)** versions that are supported for 5 years.

## 2. Nagios Core Overview

**Nagios Core** is an open-source infrastructure monitoring system developed since 1999, which enables the real-time supervision of servers, network services, applications, and IT equipment. Its strength lies in its proven reliability, its completely free nature, and its exceptional

extensibility thanks to thousands of available plugins. It offers comprehensive monitoring features (hosts, services, system resources), multi-channel alerting, and reporting with incident history. Nagios Core is distinguished by its configuration flexibility, its low resource consumption, and the support of an active community for over 20 years.

### 3. Pfsense Overview

**pfSense** is a firewall/router distribution based on FreeBSD, developed since 2004, which offers a complete network security solution with an intuitive web interface. Its main features include an advanced stateful firewall, support for multiple VPN technologies, multi-WAN routing with load balancing, QoS, VLANs, and a captive portal for WiFi authentication. Its major strength lies in its free availability, its exceptional stability inherited from FreeBSD, its modern and ergonomic web interface, and its extensible package system allowing the addition of features such as IDS/IPS (Suricata), ad-blocking, or traffic analysis.

## III. Installing and configuring Nagios core on Ubuntu Server

### 1. Prerequisites

**Minimum System Requirements**

- Processor: 1 GHz or faster (2 cores recommended)
- RAM: 1 GB minimum (2 GB or more recommended)
- Storage: 5 GB available (more depending on the number of logs, plugins, and monitored hosts)
- Network: Static IP (important for stable web access)

Operating System

- Ubuntu Server 20.04 LTS, 22.04 LTS, or 24.04 LTS
- User with sudo privileges

**System Update**

Before any installation, it is essential to ensure that the system is up to date:

```
- sudo apt update && sudo apt upgrade -y
```

**Web Server (Apache2) + PHP**

Nagios Core uses a web interface, which means that:

- Apache2 is required
- PHP is used for the interface (libapache2-mod-php)

**Nagios User Account**

A dedicated nagios user and a nagcmd group are required to run Nagios and its plugins using the commands below:

```
- sudo useradd nagios
- sudo groupadd nagcmd
- sudo usermod -a -G nagcmd nagios
- sudo usermod -a -G nagcmd www-data
```

## 2. Basic installation and configuration of Nagios core

### Installing dependencies

We first install the dependencies by executing the following command:

```
- sudo apt install -y wget build-essential apache2 php php-gd openssl perl make
  libapache2-mod-php libperl-dev libssl-dev daemon postfix mailutils unzip
```

### Downloading and compiling Nagios Core

After navigating to the tmp directory, enter the following command:

```
- wget https://github.com/NagiosEnterprises/nagioscore/releases/download/nagios-
  4.4.14/nagios-4.4.14.tar.gz
```

The downloaded file is compressed. We will therefore decompress it to enter the **nagios-4.4.14** directory and start the compilation. Command:

```
- tar xzf nagios-4.4.14.tar.gz
- cd nagios-4.4.14
- ./configure --with-command-group=nagcmd --with-
  httpdconf=/etc/apache2/sitesenabled
- make all
- sudo make install
- sudo make install-daemoninit
```

### Installing Nagios Plugins

We will retrieve the Nagios plugins from the web and install them using the following instructions:

```
- cd /tmp
- wget https://github.com/nagios-plugins/nagios-plugins/releases/download/release-
  2.4.6/nagios-plugins-2.4.6.tar.gz
- tar xzf nagios-plugins-2.4.6.tar.gz
- cd nagios-plugins-2.4.6
- ./configure --with-nagios-user=nagios --with-nagios-group=nagios --with-openssl
- make
```

```
- sudo make install
```

**Installing NRPE**

NRPE (Nagios Remote Plugin Executor) is a monitoring agent that allows Nagios (an IT infrastructure monitoring tool) to monitor resources on remote machines. To install it, follow the procedure below:

```
- cd /tmp
- wget https://github.com/NagiosEnterprises/nrpe/releases/download/nrpe-4.1.0/nrpe-4.1.0.tar.gz
- tar xzf nrpe-4.1.0.tar.gz
- cd nrpe-4.1.0
- ./configure --enable-command-args --with-ssl-lib=/usr/lib/x86_64-linux-gnu
- make all
- sudo make install
- sudo make install-config
```

**Apache Configuration**

This configuration is crucial because it enables the display of the Nagios Core web monitoring interface:

```
- sudo a2enmod rewrite cgi

- sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
```

**Contact Configuration (Notifications)**

After configuring Apache, we set the permissions: read/write for Nagios and read-only for others. Next, we managed the alert notifications by editing the file nano /usr/local/nagios/etc/objects/contacts.cfg, in which we provided the email address email_address@gmail.com where the alerts will be routed.

**Interface access test**

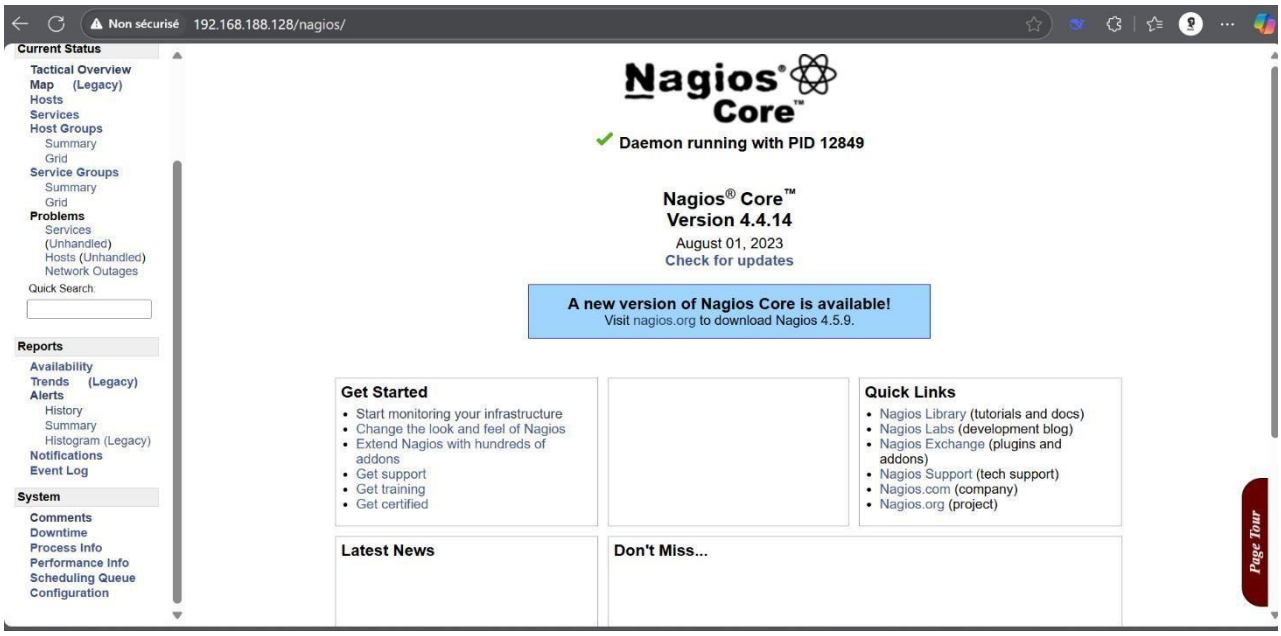


*Figure 4: Nagios core dashboard*

## IV. Implémentation de l'environnement à superviser

### 1. Architecture de supervision

The core principle of network monitoring involves a central monitoring server and one or more clients (machines, services, etc.) that are monitored by it. In our case, we are using **Nagios Core** as the monitoring server, while the clients to be monitored include a **web server**, a **file server**, and a **DHCP server**.

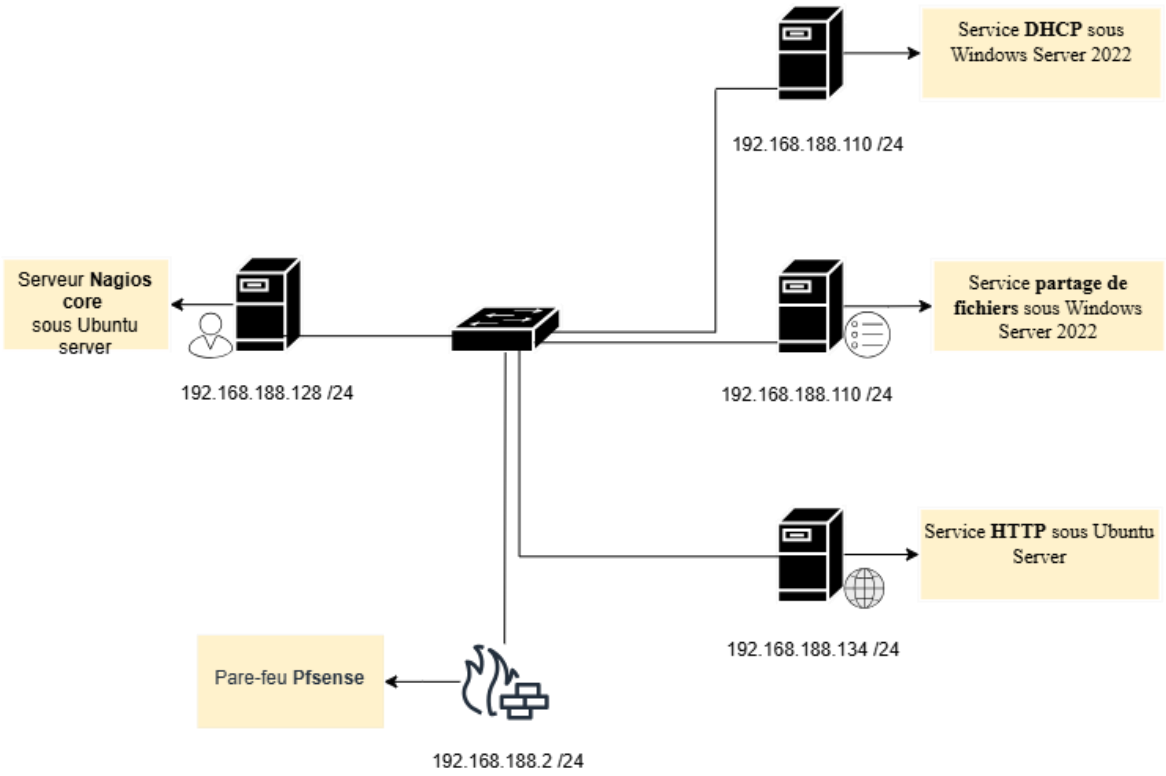


*Figure 5: Monitoring architecture*

## 2. DHCP Service Implementation and File Sharing on Windows Server 2022

### ✞ Service DHCP

**DHCP (Dynamic Host Configuration Protocol)** is a network protocol used to dynamically assign IP addresses, which avoids the need to manually assign them to each workstation one by one.

**Prerequisites:**

To prepare the DHCP service, it is essential to have certain elements already in place:

- **Windows Server 2022** installed.
- A network interface configured with a **static IP address**.
- Logged in with an **administrator account**.
- The **DHCP role** not yet installed.
- **User accounts** created.

**Installation and Configuration**

The installation of the feature is carried out through the Server Manager.

**Creating a Scope**

The scope defines the range of IP addresses that the server can assign to clients. In our callcenter setup, we use a /24 subnet mask and a 30-day lease duration, with the following address range: [192.168.188.150 - 192.168.188.250].

Configuring DHCP Options (Gateway, DNS)

Continuing through the wizard:

- Router (Default Gateway): 192.168.188.2
- DNS Servers: 192.168.188.2 (Internal DNS server)
- WINS Servers: Optional
- Click "Activate this scope now" to enable it.

**Verifying the Configuration**

On a client workstation (Windows) within the same network, execute: ipconfig /release followed by ipconfig /renew. Then, verify with ipconfig that the obtained address is within the correct range and test network connectivity using ping commands.

## ❖ File Sharing

File sharing allows users or user groups to access a shared folder over the network with specific permissions (read-only, modify, etc.), utilizing a file server—in our case, Windows Server 2022. After installing the File Server role and creating a new share on a volume, we create the shared folder on that volume. In our scenario, the path is: C:\Shares\callcenter.

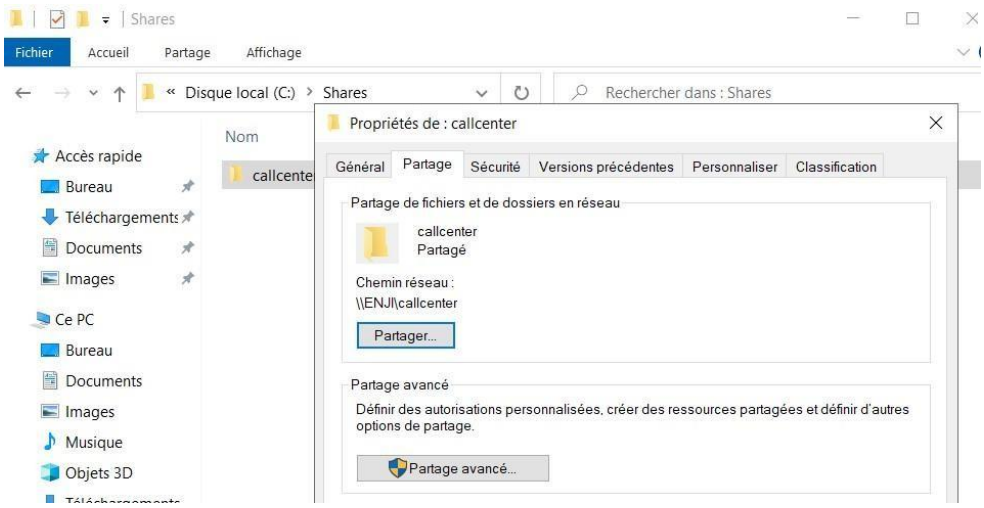


*Figure 6 : Illustration du partage du dossier*

### 3. Implémentation de l'hébergement d'un site web sous Ubuntu server

After installing Ubuntu server on our VMware workstation hypervisor and doing the updates, I proceeded to install the Apache web server by running:

- *apt install apache2 -y*
- *systemctl status apache2*

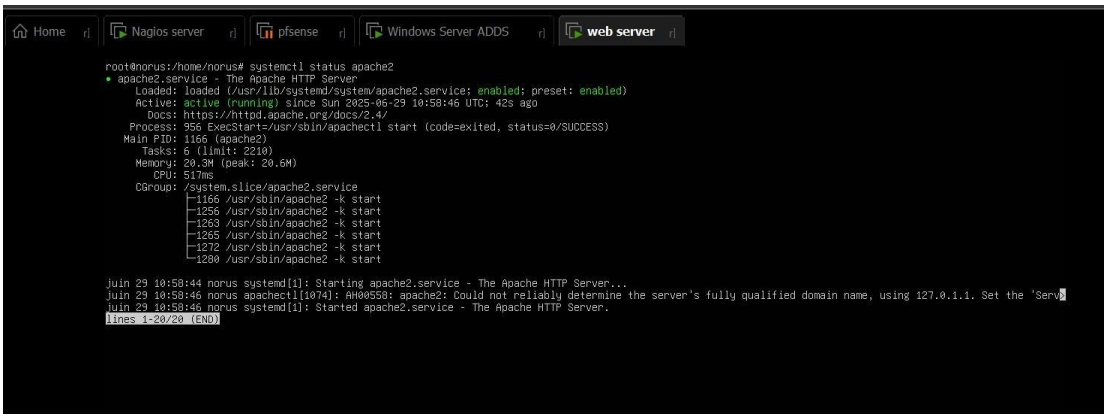


*Figure 7 : Serveur Apache installé*

**Creating the folder to host our website**

- *mkdir -p /var/www/site_web/web_site*

**Setting permissions**

- *chown -R $USER:$USER /var/www/site_web/web_site*
- *chmod -R 755 /var/www/site_web*

After configuration and hosting, we access our website via our web browser by typing in the search bar: **http://192.168.188.134/call-center.html**

*Figure 8 : Page web callcenter*

## 4. Integration of servers and services to be monitored on Nagios core

### Configuring Nagios for the web server

On the Nagios server, we had created the configuration directory:

```
- mkdir /usr/local/nagios/etc/servers
```

Then edited the main file by adding **cfg_dir=/usr/local/nagios/etc/servers** to line 51 in order to tell Nagios to read all the .cfg files in the servers directory, with the command:

```
- sudo nano /usr/local/nagios/etc/nagios.cfg
```

### Creating the web server configuration:

It was by modifying the nano file **/usr/local/nagios/etc/servers/webserver.cfg** that we created the configuration of the web server. We had indeed filled in the necessary parameters to allow Nagios to supervise our web server.

### Nagios Configuration for Windows Server 2022

First of all, we had installed and configured NSclient++ on windows server. Then on Nagios:

**Creating the configuration file:**

It was by modifying the nano file **/usr/local/nagios/etc/servers/windows-server.cfg** that we created the web server configuration. We had indeed filled in the necessary parameters to allow Nagios to supervise our web server.

**Verification at the web interface level:**

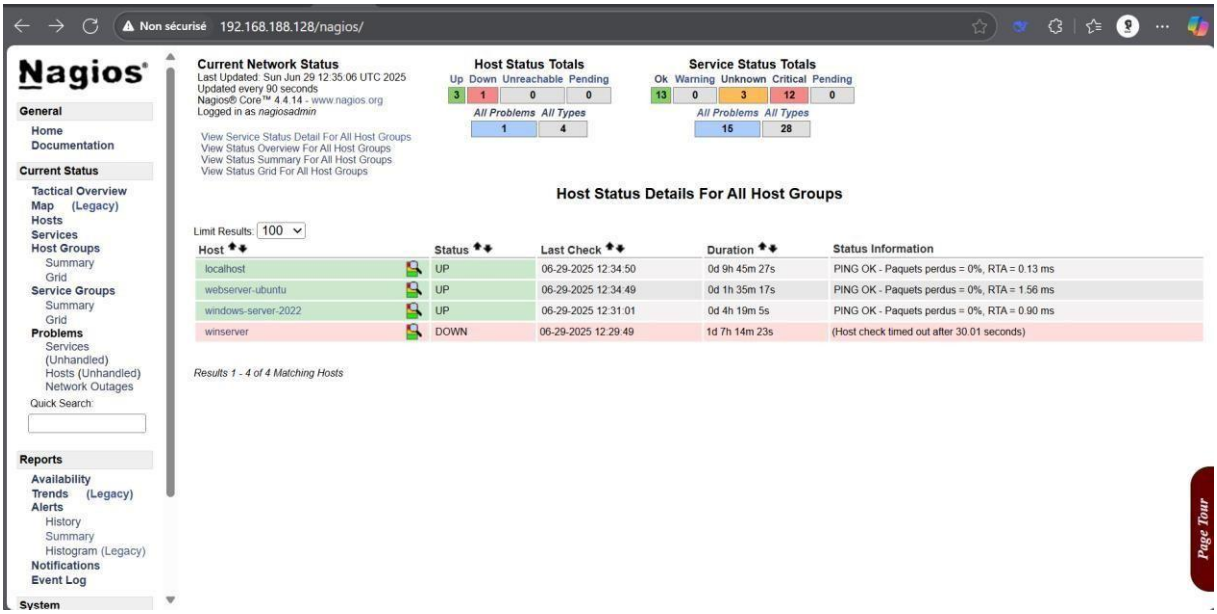


*Figure 9: Windows Server and Web Server are running*

### 5. Filtering the data flow in the LAN with Pfsense

Pfsense comes with a default rule that blocks incoming traffic into the LAN. And we created a rule that allows the LAN to go over the internet, but only through the firewall interface, which has become its default gateway.

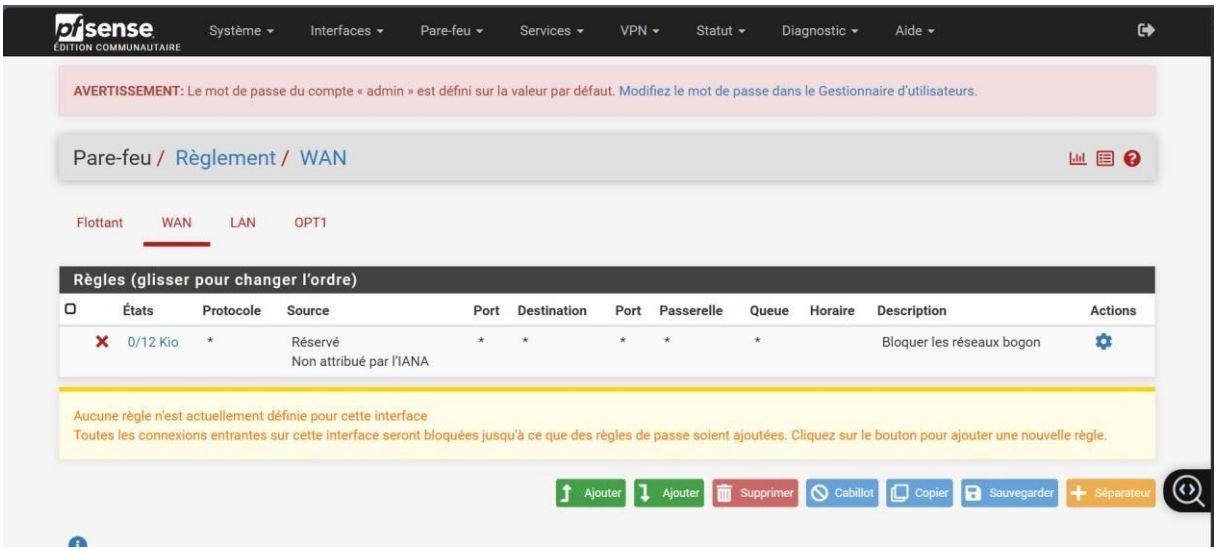


*Figure 10: WAN Filter Rule*

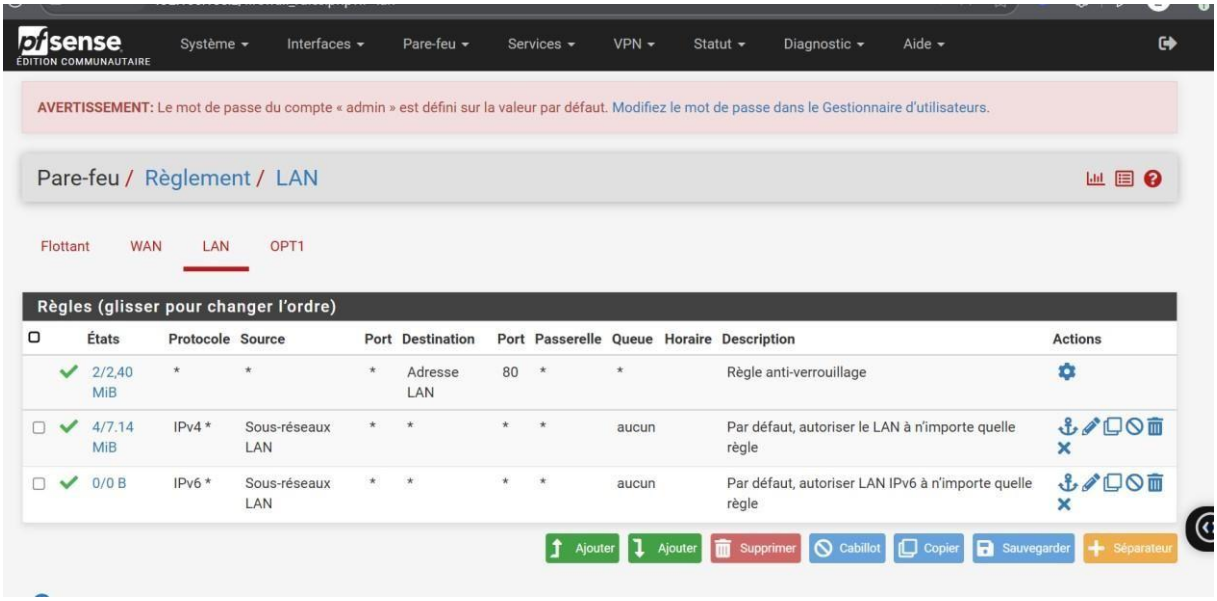


*Figure 11: LAN Filter Rule*

## Section 2: Interpretation and validation of results

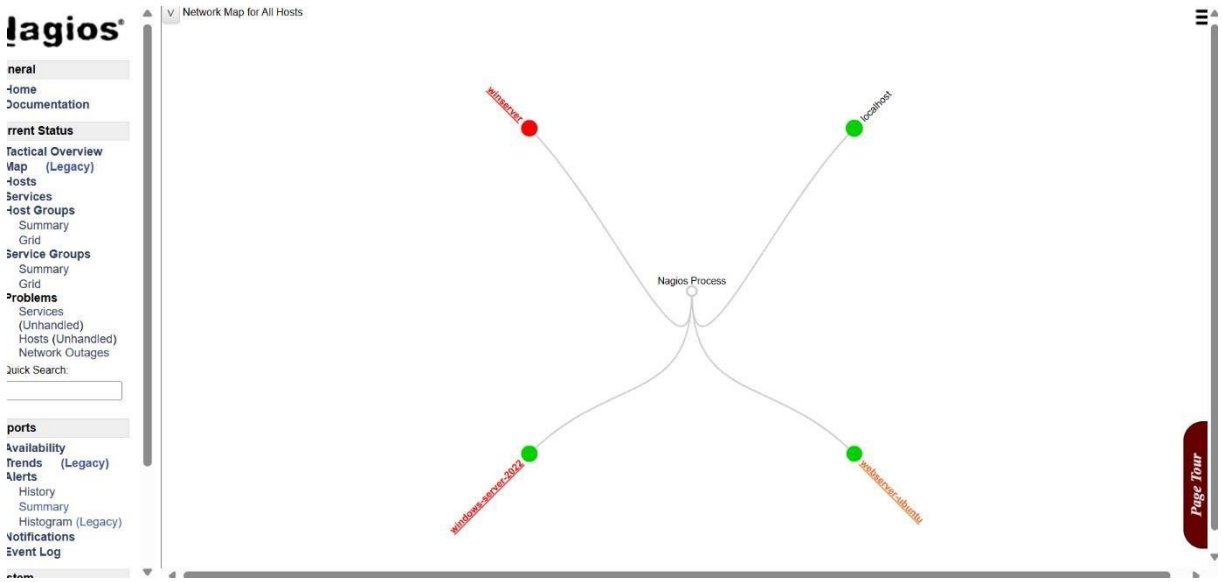


*Figure 12: Nagios Monitored Host Mapping*

We can see there, the mapping of the hosts supervised by Nagios Core where we find our two windows server 2022 servers and the web server (webserver-ubuntu) which testifies to the consideration of the configurations that aimed to monitor these two hosts.

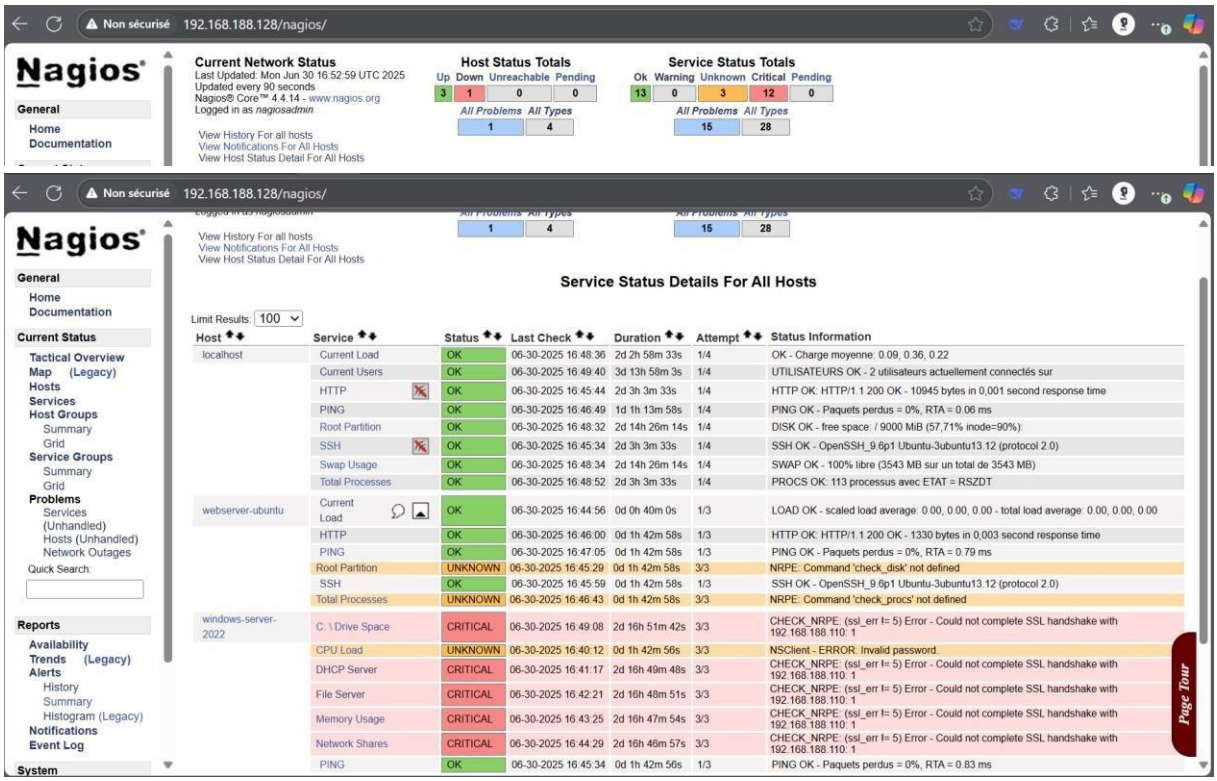


*Figure 13: Nagios Service Oversight*

This image represents the Nagios Core supervision screen where we see three main machines: localhost, webserver-ubuntu and windows-server-2022. For each machine, Nagios displays the status of the services it monitors such as ping, disk space, HTTP services, SSH, etc. Colors help you quickly spot what's going well **(green),** what's problematic **(red),** what's unknown **(gray or yellow),** and what's pending (light blue).

**Machine1** : localhost

This is the machine that Nagios is installed on, so it monitors itself. Here's what we can understand:

1. The system load is good. This means that the server is not overloaded and is working normally.
2. The number of users logged in is correct. There are two people connected, it's reasonable.
1. The HTTP server is marked in red (CRITICAL). This means that he responds too slowly. Generally, this kind of slowness can come from a slow service or a lack of resources.

1. Ping works great. This confirms that the machine is accessible on the network.
2. The main disk (root) has enough free space (75% available), so no risk of being full.

3. The SSH service is active. So you can connect to this machine remotely.
4. The number of processes is normal. There aren't too many programs running at the same time.

In summary, the main server is fine, except for the HTTP service which needs to be checked or restarted.

**Machine 2** : webserver-ubuntu

This is our web server under Ubuntu. The analysis states:

1. The load test is in progress, but not yet completed (PENDING).
2. HTTP service responds smoothly
3. Pings also go through without a problem.
4. The disk space test says unknown, as does the process test. This comes from the fact that the NRPE service (which allows Nagios to ask questions to Ubuntu) does not work as it says, will have to review the configurations.

All in all, the Ubuntu server doesn't work well but it is shown that it is already quite operational. However, the configuration of the NRPE service must be reviewed.

Machine 3 : **windows-server-2022**

This is the server on which the DHCP and file server roles are installed. Here's what Nagios says:

5. The disk space test on C: fails. Nagios can't connect.
6. The DHCP server, file server, memory used, and network share are all in critical error. There is an SSL communication error (failed handshake). Basically, Nagios and the server can't talk to each other because of an SSL-related problem.
7. For the DHCP server, there is also a password error: the SNMP password is not correct.
8. On the other hand, the ping works. This shows that the machine is well turned on and accessible.

In short, the Windows server is online, but everything related to SNMP monitoring fails.

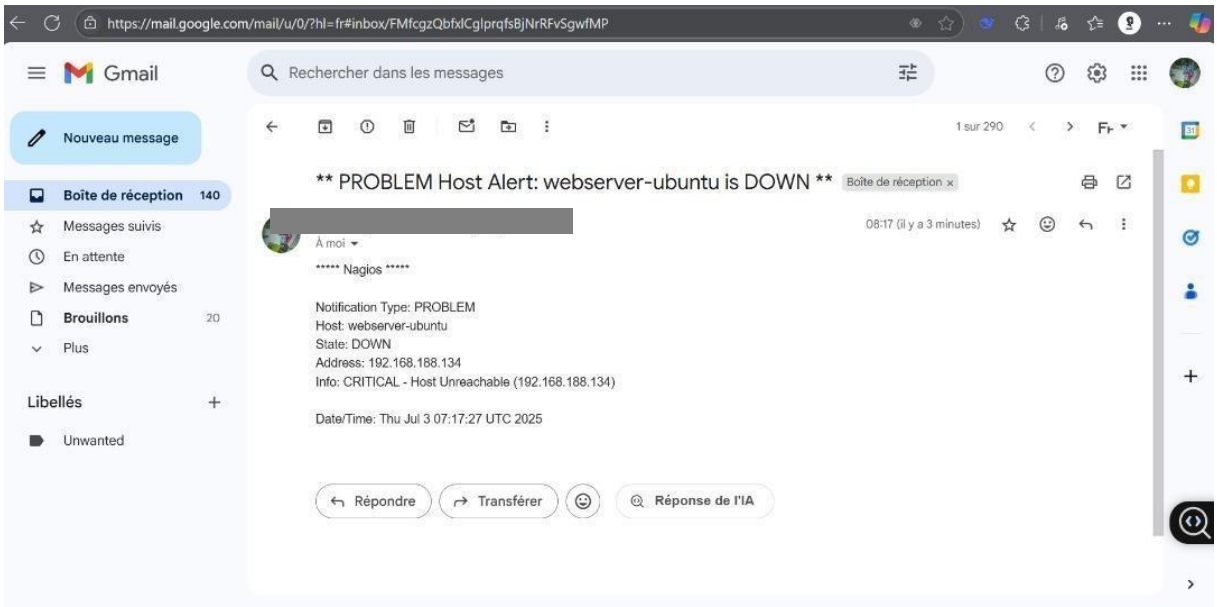


*Figure 14 : Notification d'une alerte par Nagios*

Cette image montre que l'envoi des alertes à l'adresse mail que nous avions rentré en amont par Nagios fonctionne parfaitement.

## Conclusion

At the end of this project to deploy a network monitoring solution on Linux, several major achievements emerged:

1. **Achievement of objectives**

It was a question of setting up a virtual infrastructure (VMware Workstation) including the Nagios Core supervisor, two servers (Ubuntu Server for the web service, Windows Server 2022 for DHCP and file services) and a pfSense firewall. Configure critical services (DHCP, file sharing, HTTP, SSH, system resources) for reliable and responsive monitoring. The functional Nagios interface allows you to view the status of hosts and services (colors, severity levels).

2. **Skills developed**

    1. Proficiency in virtualization tools (VMware), Linux systems and Windows Server.
    2. Getting started with open source monitoring technologies (Nagios, plugins, verification scripting).
    3. Network security management via secure SNMP, firewall configuration, and auditing.

3. **Answer to the problem**

The challenge was to monitor the state of the network in real time, anticipate failures and guarantee high availability of services. The solution implemented effectively meets these challenges:

1. **Real-time monitoring** : Nagios provides continuous monitoring of hosts and services with regular checks.
2. **Anomaly detection** : The system automatically identifies faults and malfunctions
3. **Proactive notifications** : Email alerts allow for quick admin response
4. **Centralized interface** : The web dashboard provides a holistic view of the infrastructure

1. **Hypothesis validation**

The hypothesis that a properly deployed and configured open-source network monitoring solution will enable effective real-time monitoring is partially validated:

1. **Points validated:**
    1. The open source Nagios Core solution is indeed capable of providing effective monitoring
    2. Deployment costs are minimal (no license required)
    3. Responsiveness in the event of an anomaly is ensured by the alert system
    4. The impact on the functioning of the organization is minimal
2. **Points to be qualified:**
    1. Configuration remains complex and time-consuming, especially for Windows environments
    2. Some communication issues (SNMP, SSL) limit the effectiveness of monitoring

- The user interface, while functional, lacks modernity